Yabetse Tesfaye Alemu

Software Engineer : Backend, ML Infrastructure & Systems : 6+ Years Experience

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PROFESSIONAL EXPERIENCE

Revelo : AI Infrastructure & DevOps Training Engineer — 03/2026 – 05/2026 · Remote

- Created and evaluated complex Infrastructure-as-Code tasks in **Terraform, YAML, JSON, Python, and AWS CDK** to improve AI training and evaluation for Amazon Nova Lite v2.
- Designed high-difficulty prompts and validation scenarios benchmarking model reasoning on real-world cloud infrastructure problems.

Upwork : Mobile Application Developer, FlutterFlow (Freelance) — 12/2024 – 03/2026 · Remote

- Delivered cross-platform mobile applications for international clients using **Flutter and FlutterFlow**, from program creation through release.

Eskalate : Full Stack Engineer (Full-time) — 01/2021 – 09/2024 · On-site

- Engineered scalable, reusable **authentication and authorization systems**, enhancing platform security and increasing feature development efficiency by 30%.

Eskalate : Full Stack Engineer (Part-time) — 11/2019 – 01/2021 · Hybrid

- Built and improved authentication and authorization systems, helping create a more secure and reliable application.

PROJECTS

RustV OS Lab Bare-Metal RISC-V Operating System

A no-std Rust kernel booting on QEMU without BIOS or SBI: Sv39 virtual memory, preemptive scheduling, user-mode syscalls, a from-scratch virtio-MMIO DMA driver, and TinyFS storage.

Rust · RISC-V Assembly · QEMU · Sv39 VM · Virtio-MMIO · TinyFS

GAN Observatory Real-Time GAN Training Observatory

Streams GAN training telemetry to a browser dashboard at ~20 fps via FastAPI WebSockets; from-scratch NumPy GAN with live hyperparameter updates and mode-collapse detection.

Python · NumPy · TensorFlow/Keras · FastAPI · WebSockets · Docker

ImmuStore DB Distributed Database Engine Built from Scratch

Pure-Python distributed database: copy-on-write B+ tree storage, crash-safe MVCC via shadow-paging, a Redis-compatible RESP asyncio server, and from-scratch Raft replication with automatic failover.

Python · asyncio · B+ Trees · MVCC · Raft · Redis RESP

Gebeta Traditional Ethiopian Mancala Game with AI

Gebeta game for desktop and web (Java Swing + TypeScript) with local multiplayer and a time-bounded AI using iterative-deepening minimax with alpha-beta pruning.

Java · TypeScript · Minimax · Alpha-Beta · Gradle · Vite

AI-RAD Healthcare Screening & Records Platform

Hospital COVID-19 screening and digital records with role-based access. Won **1st place** in the 2020 UNDP & MINT COVID-19 Innovation Challenge (580,000 ETB grant).

React · Node.js · Firebase · Firestore · RBAC

SKILLS

Languages	Python, Rust, Go, Java, C#, TypeScript, JavaScript, Dart, SQL, RISC-V Assembly
Backend	FastAPI, Node.js, SQLAlchemy, REST, WebSockets, asyncio, Celery, Redis, Firebase, SQLite, MongoDB
ML & Systems	TensorFlow/Keras, scikit-learn, NumPy, GANs, distributed systems, Raft, MVCC, OS internals
Frontend	React, Next.js, Flutter, Unity, Java Swing, Tailwind CSS
DevOps	Docker, Kubernetes, Git, GitHub Actions, CI/CD, pytest, JUnit 5, Ruff, Gradle, Vite, QEMU, AWS/GCP

EDUCATION

BSc Software Engineering

ACT American College of Technology

2014 – 2019 · Addis Ababa, Ethiopia

Foundation in DS & algorithms, databases, OS, networks, and software architecture through full-lifecycle team projects.

Software Engineering Program : DS & Algorithms

A2SV : Africa to Silicon Valley

2019 – 2020 · Addis Ababa, Ethiopia

Intensive algorithmic problem solving and interview prep with complexity analysis and peer code reviews.